

MECH 151: Technical Design – Prototyping

Prototype Design Proposal

*Final Exam — Prototype Project (team submission)***Team Name / Number:** _____**Team Member 1:** _____**Team Member 2 (if applicable):** _____**Team Member 3 (if applicable):** _____**Date:** _____

What This Proposal Establishes

This proposal establishes your team, what you're proposing to build, and how your design will meet the Final Exam's two required elements: integrating at least 2 different bulk material types, and demonstrating at least 1 joining method between 2 parts. Proposals are reviewed during the Week 13 design review/iteration workshop — expect feedback and a possible revision before you begin building.

Design Concept

What are you proposing to build? What problem does it solve or what function does it serve? Include a rough sketch or reference image if helpful.

Bulk Material Integration (required: at least 2 types)

List each bulk material type your design uses. A design must integrate at least 2 different types (e.g., thermoplastic + raw/processed stock).

Material #	Material Type	Used In / For	Why This Material
1			
2			

Add more rows if your design uses additional materials.

Joining Method (required: at least 1, between 2 parts)

Which joining method will you use (e.g., heat-set threaded insert, snap fit, press/interference fit, compliant/living hinge, threaded fastener)? Between which two parts? Why did you choose it over other options covered in Unit 3?

Manufacturing Plan

Which shop process(es) will you use to build this (additive/3D printing, subtractive/CNC, laser cutting)? What is your rough build timeline?

Proposal Review Notes

For instructor use during the Week 13 design review/iteration workshop.

Feedback:

Proposal Status (Approved / Revise & Resubmit): _____